

Workshop

K-means Clustering

$$1. C_i = \{ x \in D \mid \arg \min_{j \in [1, k]} d(x, m_j) = i \}$$

$$2. m_i = \frac{\sum_{x \in C_i} x}{|C_i|}$$

Where is the distance function used?

DBSCAN: Density-based Clustering

1. Estimate density for each point in a given dataset
 ϵ -Neighborhood Density Estimator is used
2. Identify all high-density points having density more than τ
3. Link neighboring high-density points to a cluster

Where is the distance function used?

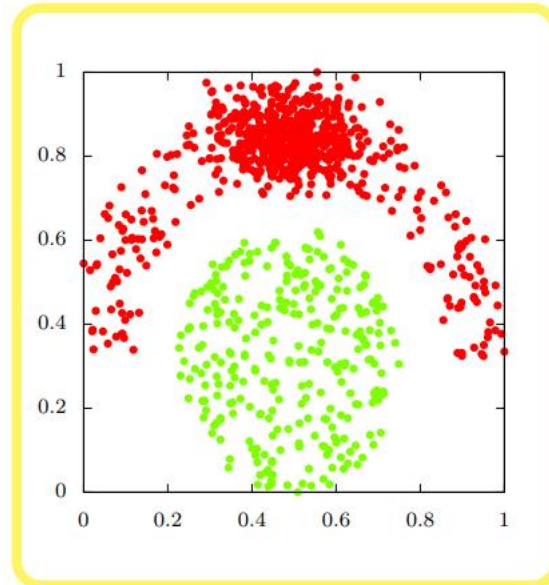
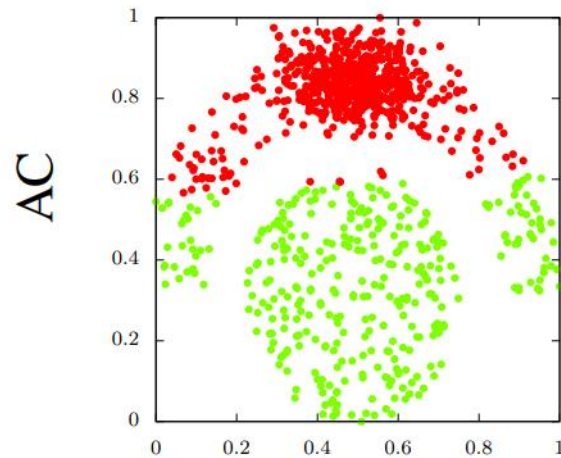
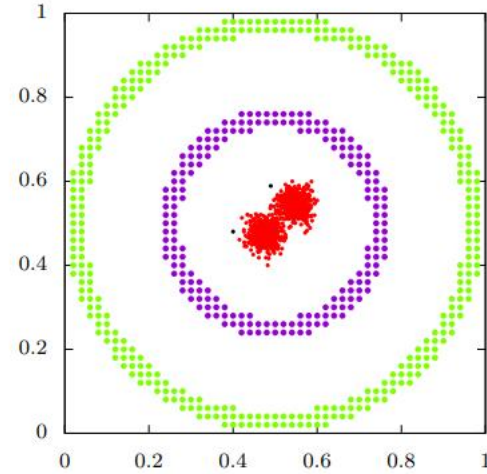
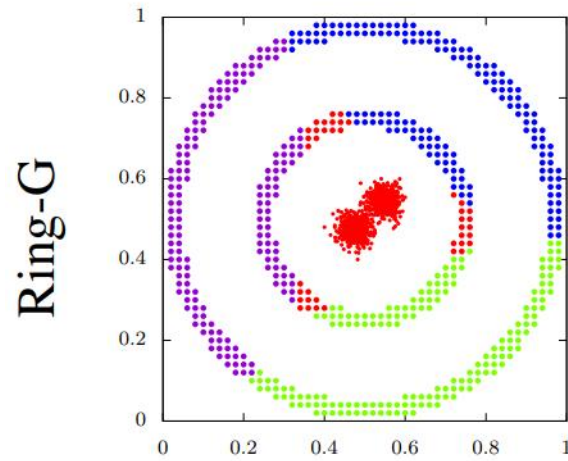
Data independent distance/similarity measures

Purely based on the geometry position of the points in the data/input space, independent of data distribution.

- L_p Norm (Minkowski distance): $Dist(\mathbf{X}, \mathbf{Y}) = \left(\sum_{i=1}^d |x_i - y_i|^p \right)^{1/p}$
- Cosine similarity: $\text{cosine}(\mathbf{X}_1, \mathbf{X}_2) = \langle \mathbf{X}_1, \mathbf{X}_2 \rangle / \|\mathbf{X}_1\| \|\mathbf{X}_2\|$
- Kernel functions: Gaussian kernel, Laplacian kernel, exponential kernel

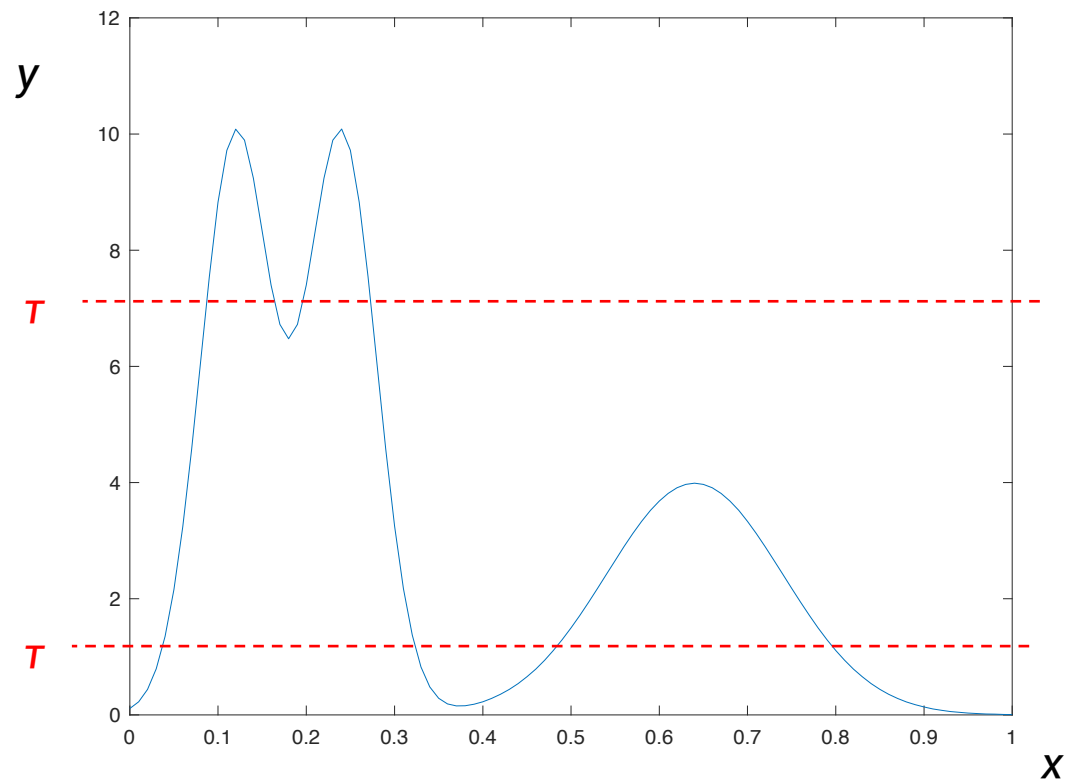
$$\text{Gaussian kernel: } \kappa(x, y) = \frac{1}{(\sigma \sqrt{2\pi})^d} \exp \left(-\frac{\|x - y\|^2}{2\sigma^2} \right).$$

1. Why does k-means clustering produce these clustering outcomes?
2. Why does DBSCAN produce these outcomes?



DBSCAN: Varying density issue

$$y = pdf(x)$$



What is Isolation Kernel?

Definition 1. Isolation Kernel of any two points $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$ is defined to be the expectation taken over the probability distribution $P_{\mathbb{H}_\psi(D)}$ on all partitionings $H \in \mathbb{H}_\psi(D)$ that both $\mathbf{x}$ and $\mathbf{y}$ fall into the same isolating partition $\theta[\mathbf{z}] \in H$:

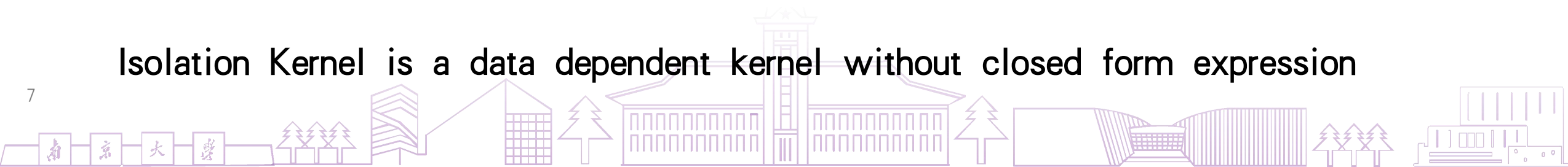
$$K_\psi(\mathbf{x}, \mathbf{y} \mid D) = \mathbb{E}_{H \sim P_{\mathbb{H}_\psi(D)}} [\mathbb{1}(\mathbf{x}, \mathbf{y} \in \theta[\mathbf{z}] \mid \theta[\mathbf{z}] \in H)], \quad (1)$$

where $\mathbb{1}(\cdot)$ is an indicator function.

Definition 2. Isolation Kernel K_ψ has a feature map $\Phi : \mathbf{x} \rightarrow \{0, 1\}^{t \times \psi}$. K_ψ has no closed form expression and is expressed in terms of Φ as follows:

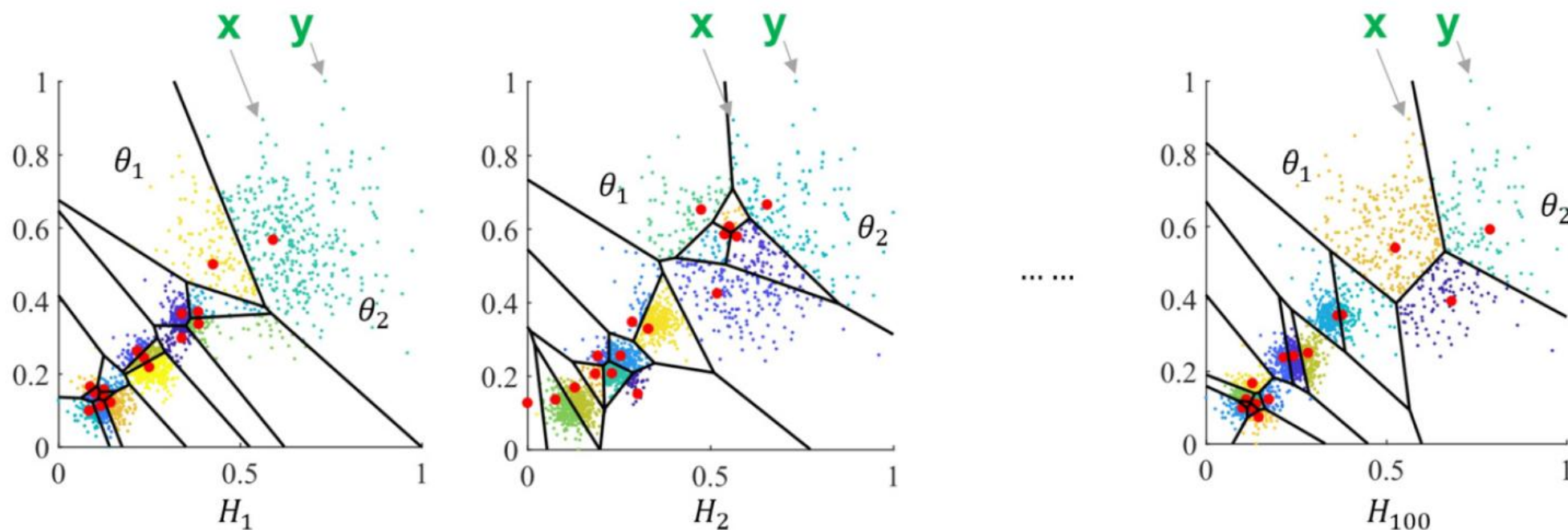
$$K_\psi(\mathbf{x}, \mathbf{y} \mid D) \simeq \frac{1}{t} \langle \Phi(\mathbf{x}|D), \Phi(\mathbf{y}|D) \rangle.$$

Isolation Kernel is a data dependent kernel without closed form expression



IK implemented using Voronoi Diagram

$$K_{\psi}(\mathbf{x}, \mathbf{y} \mid D) = \mathbb{E}_{H \sim P_{H_{\psi}(D)}} [\mathbb{1}(\mathbf{x}, \mathbf{y} \in \theta[\mathbf{z}] \mid \theta[\mathbf{z}] \in H)]$$

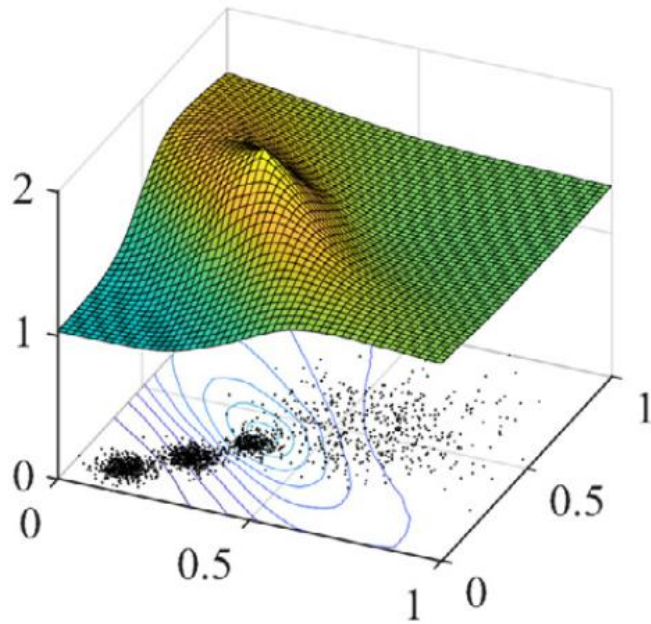


$\psi = 16, t = 100$

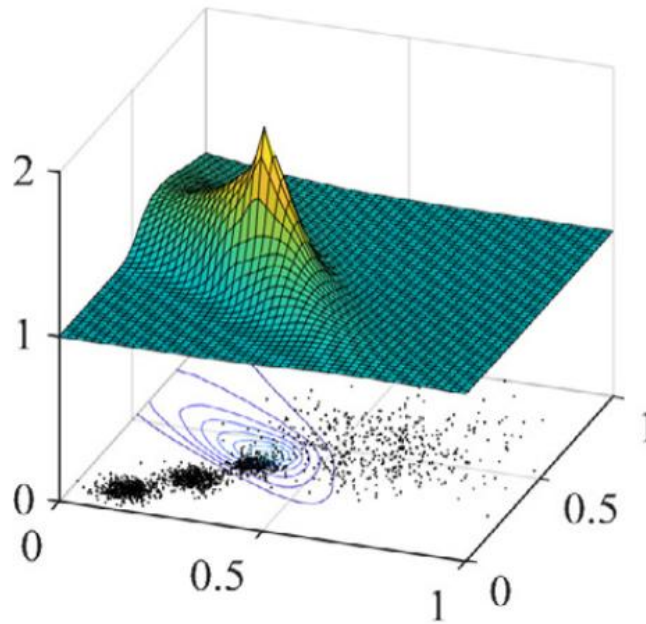


Kernel Distribution of Isolation Kernel

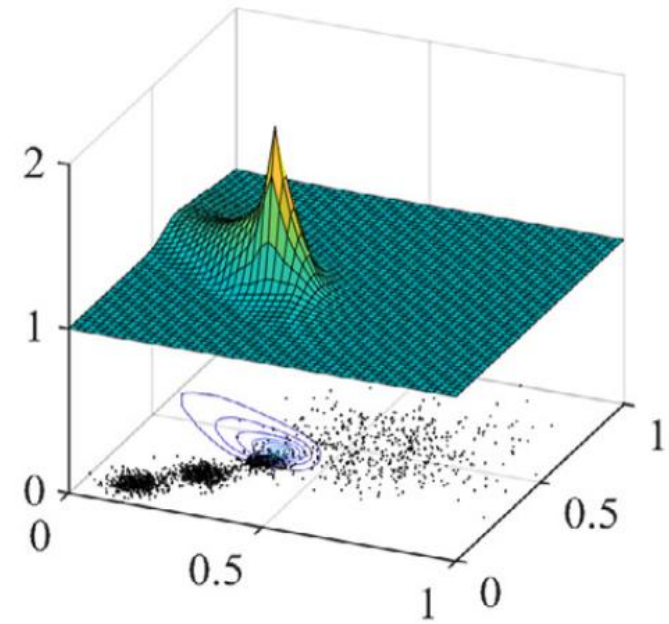
$$K_{\psi}(\mathbf{x}, \mathbf{y} \mid D) \simeq \frac{1}{t} \langle \Phi(\mathbf{x} \mid D), \Phi(\mathbf{y} \mid D) \rangle$$



(a) $\psi = 4$



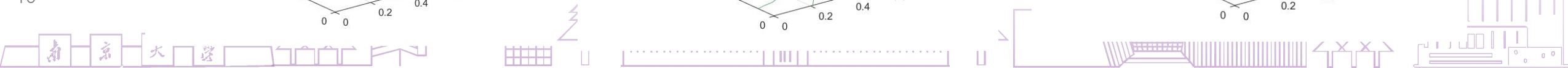
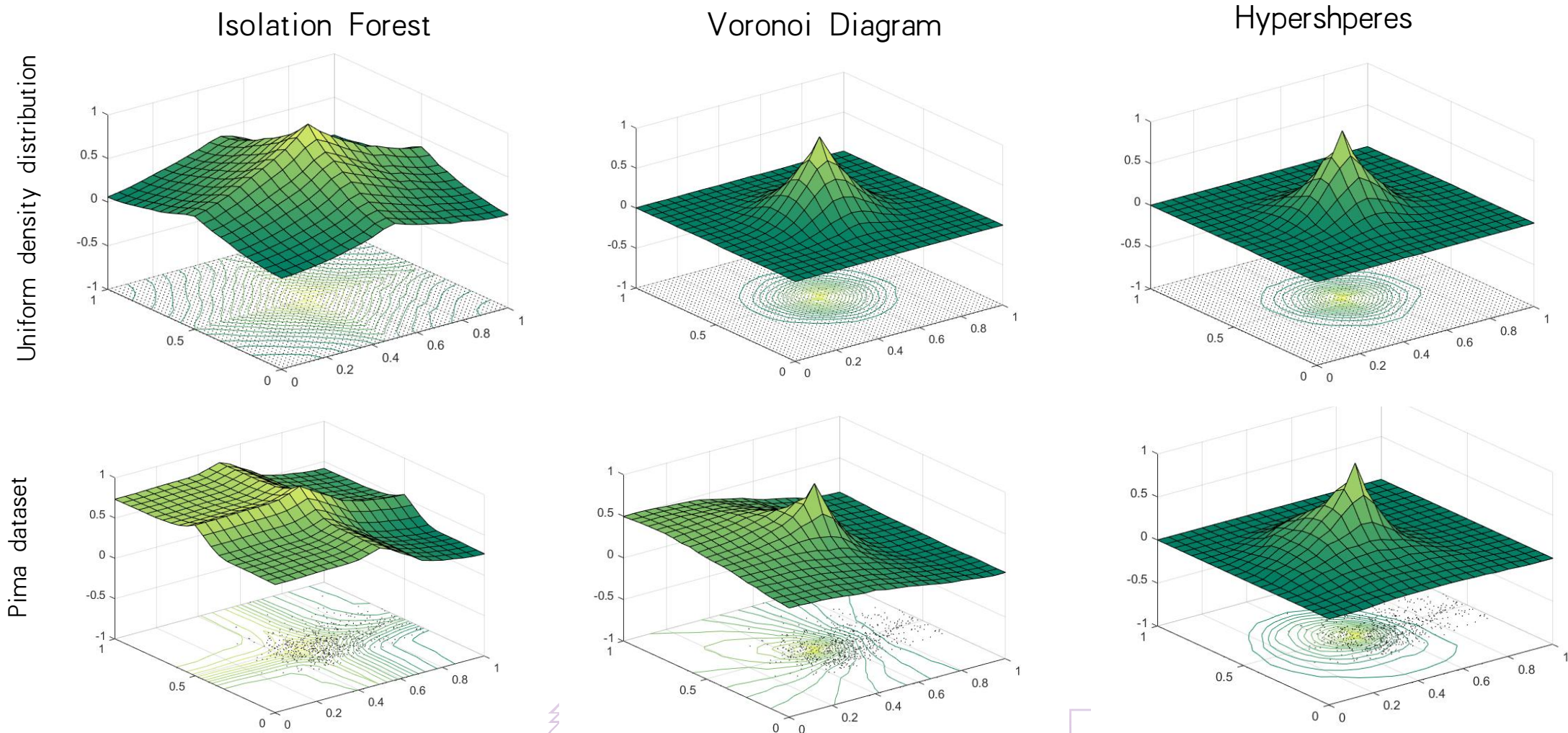
(b) $\psi = 16$



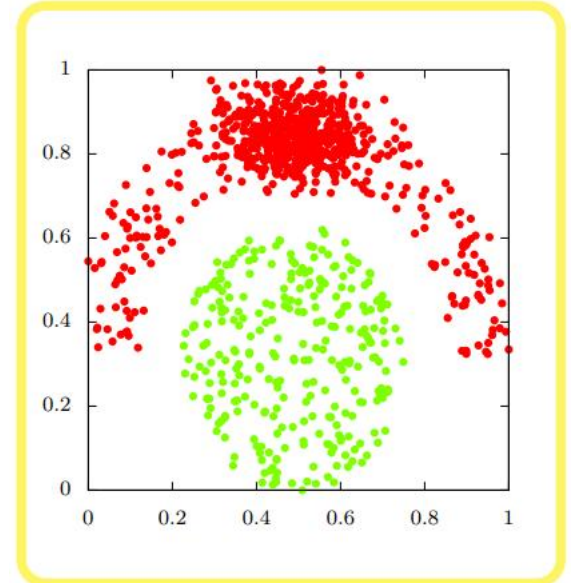
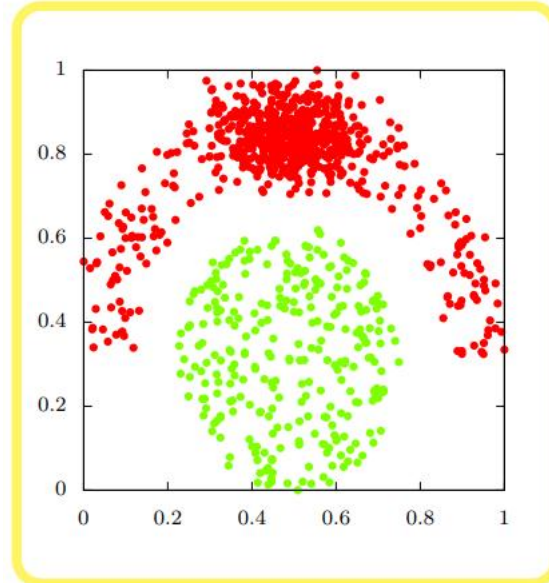
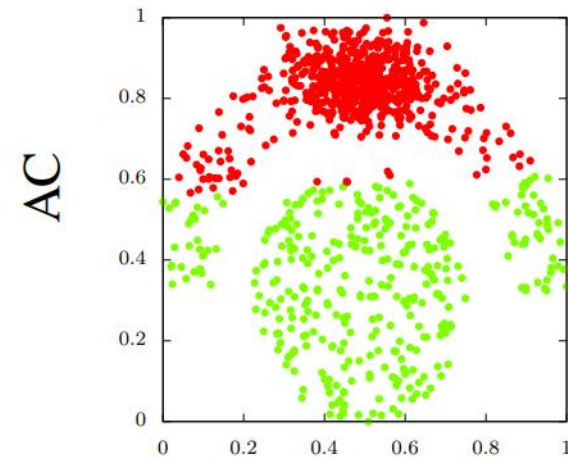
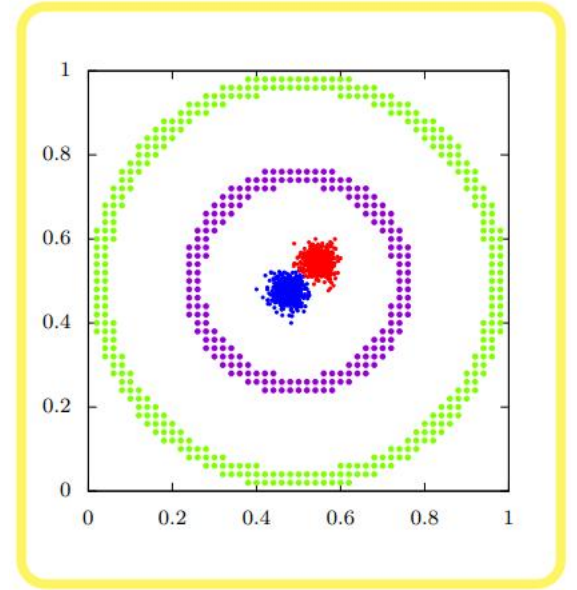
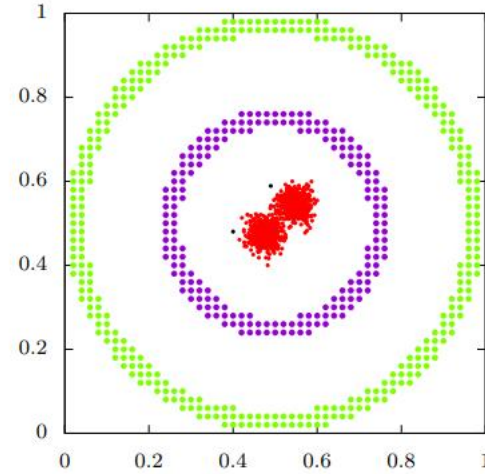
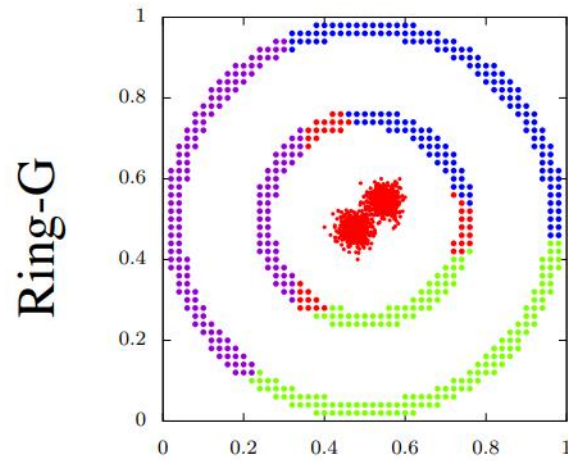
(c) $\psi = 32$



Example kernel distributions of IK of three different implementations



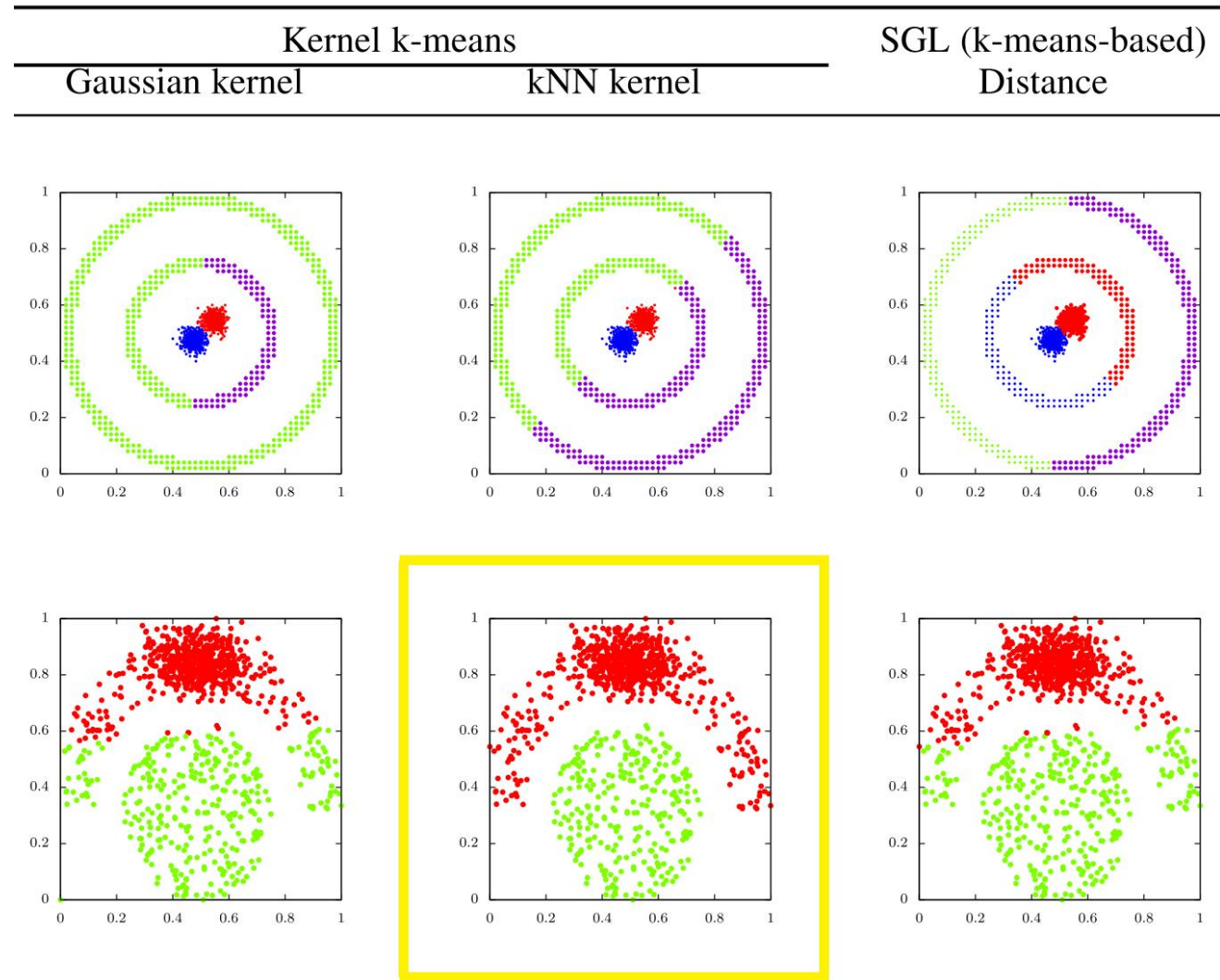
1. The algorithm and distance function used have an impact
2. **Data-dependent measure** can better deal with varied densities than **data-independent measure**



Discussion

1. Kernel has been used as to means to convert a linear model to a nonlinear model (e.g., Linear classifier to SVM classifier). This has been applied to k-means clustering to create kernel k-means clustering.

Discuss why this does not work.



Discussion

- What are the lessons learned?

The Impact of high dimensions on similarity measures

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Reading

Required reading:

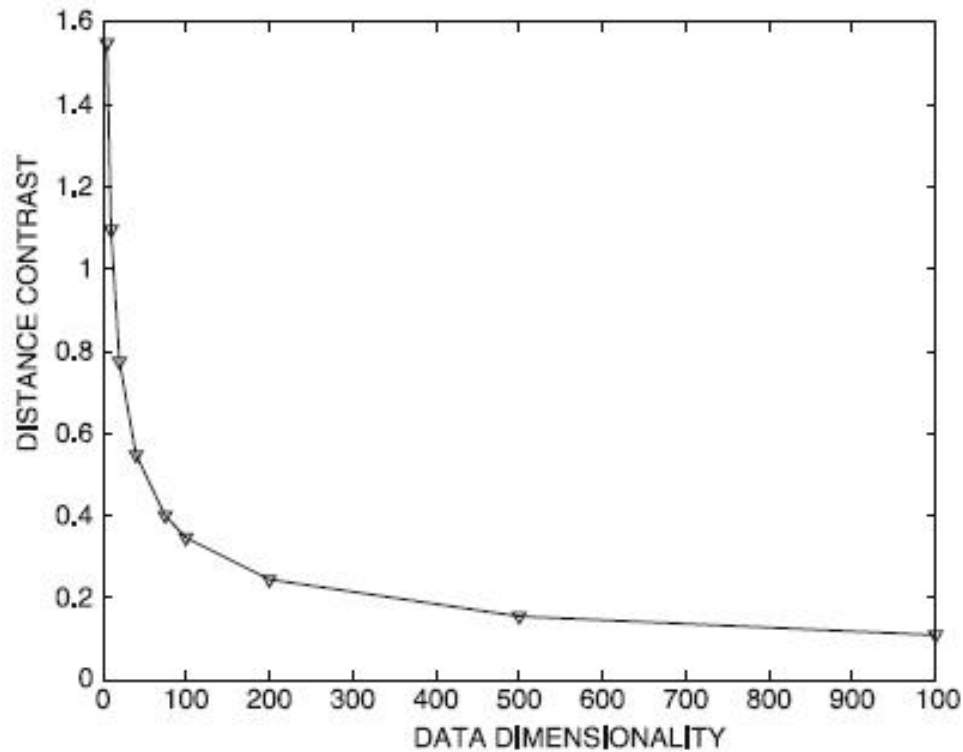
- Aggarwal Chapter 3
- Papers listed on the reading slide

Acknowledgement:

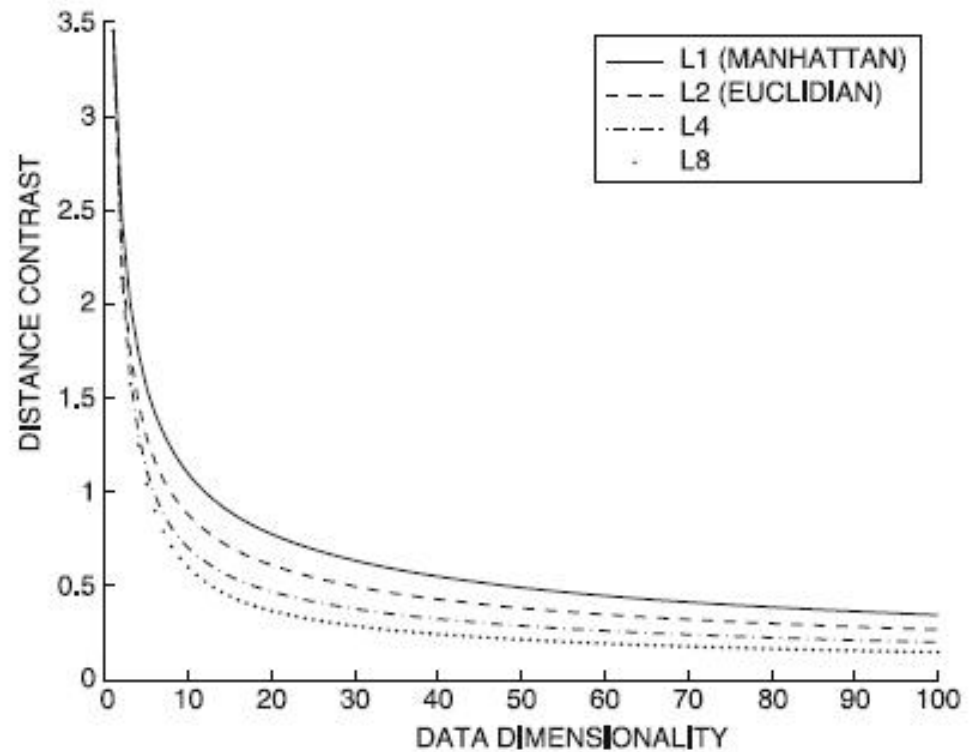
The contents and images on slides 1–31 (unless stated otherwise) are sourced from C. C. Aggarwal, Data Mining: The Textbook. Springer, 2015.

Effect of high dimensionality

$$\lim_{d \rightarrow \infty} \frac{MaxDist - MinDist}{MinDist} = 0$$



(a) Contrasts with dimensionality



(b) Contrasts with norms

Concentration of Measure

- Talagrand [2] summarizes the intuition behind the concentration of measure as:
“a random variable that depends (in a ‘smooth’ way) on the influence of many independent random variables (but not too much on any of them) is essentially constant.”
- In the context of a high dimensional space, a **distance measure** which depends on many independent dimensions (but not too much on any of them) is essentially constant.
- A broader implication on the blessing of dimensionality [1] refers to the “constant” nature of many independent dimensions [2] which can be exploited to aid mathematical analysis in high dimensions, even though the analysis is hard in a low dimensional context where the “constant” nature is non-existence.

[1] Donoho, High-Dimensional Data Analysis: The Curses and Blessings of Dimensionality, Invited Lecture at Mathematical Challenges of the 21st Century, AMS National Meeting, 2000.

[2] Talagrand, A new look at independence, Ann. Probab. 24 (1) (1996) 1–34.

The Curse and Blessing of Dimensionality

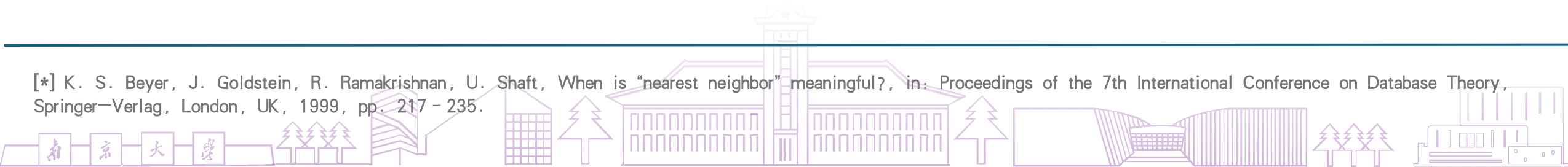
- In the last few decades, a significant amount of research has been invested in studying the curse and blessing of dimensionality of a distance measure. We now have a good understanding in terms of (i) concentration of measure and its consequence of being unable to find the exact nearest neighbor of a query, and (ii) hubness.
- Despite this advancement, breaking the curse is still elusive. A key question remains open:
“Is it possible to find the single nearest neighbor of a query in high dimensions?”



Is it possible to find the single nearest neighbor of a query in high dimensions?

- The answer to this question is negative, based on the current analyses on distance.
- Rather than finding the single nearest neighbor of a query, the issue has been diverted to finding all data points belonging to the same cluster as the query, in a distribution with well-separated clusters.
- These nearest neighbors are ‘meaningful’ for the purpose of separating different clusters in a dataset [*].
- But the (single) nearest neighbor, in the true sense of the term, cannot be obtained using a distance measure, even in the special distribution of well-separated clusters described above.

[*] K. S. Beyer, J. Goldstein, R. Ramakrishnan, U. Shaft, When is “nearest neighbor” meaningful?, in: Proceedings of the 7th International Conference on Database Theory, Springer-Verlag, London, UK, 1999, pp. 217 – 235.

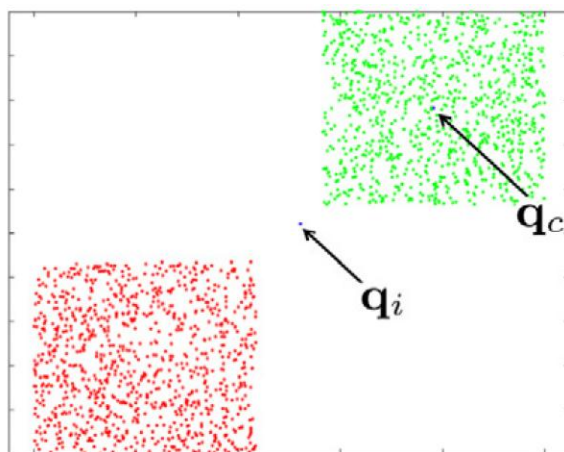


Gaussian, Adaptive Gaussian & Isolation Kernels

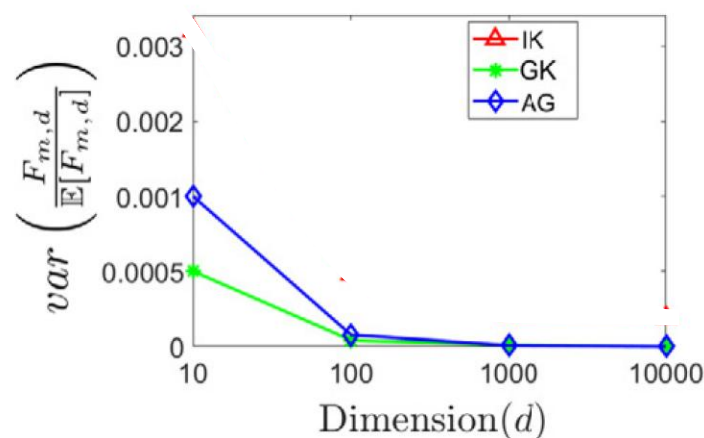
Is it possible to find the single nearest neighbor of a query in high dimensions?

No!

Due to the curse of dimensionality & concentration of measure



(a) Dataset used



(b) (In)stability of a measure

As the number of dimensions increases, the distance between any two points approaches zero [*].

[*] K. S. Beyer, J. Goldstein, R. Ramakrishnan, U. Shaft, When is “nearest neighbor” meaningful?, in: Proceedings of the 7th International Conference on Database Theory, Springer-Verlag, London, UK, 1999, pp. 217 – 235.



Instability of a distance measure [*]

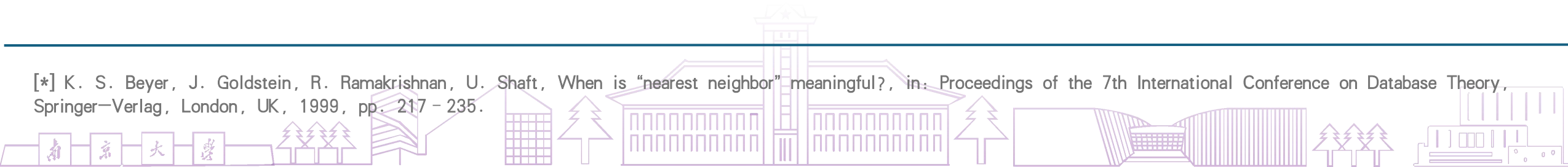
Given a dataset $\mathcal{D} = \{x_1, x_2, \dots, x_n\} \subseteq \mathbb{R}^d$ and a query point q , the nearest neighbor is traditionally defined as:

$$\text{NN}(q) = \arg \min_{x_i \in \mathcal{D}} \text{dist}(q, x_i)$$

However, in high-dimensional spaces, due to the distance concentration effect, the following situation arises:

$$\lim_{d \rightarrow \infty} \frac{\max_{x_i \in \mathcal{D}} \text{dist}(q, x_i) - \min_{x_j \in \mathcal{D}} \text{dist}(q, x_j)}{\min_{x_j \in \mathcal{D}} \text{dist}(q, x_j)} \approx 0$$

[*] K. S. Beyer, J. Goldstein, R. Ramakrishnan, U. Shaft, When is “nearest neighbor” meaningful?, in: Proceedings of the 7th International Conference on Database Theory, Springer-Verlag, London, UK, 1999, pp. 217 – 235.



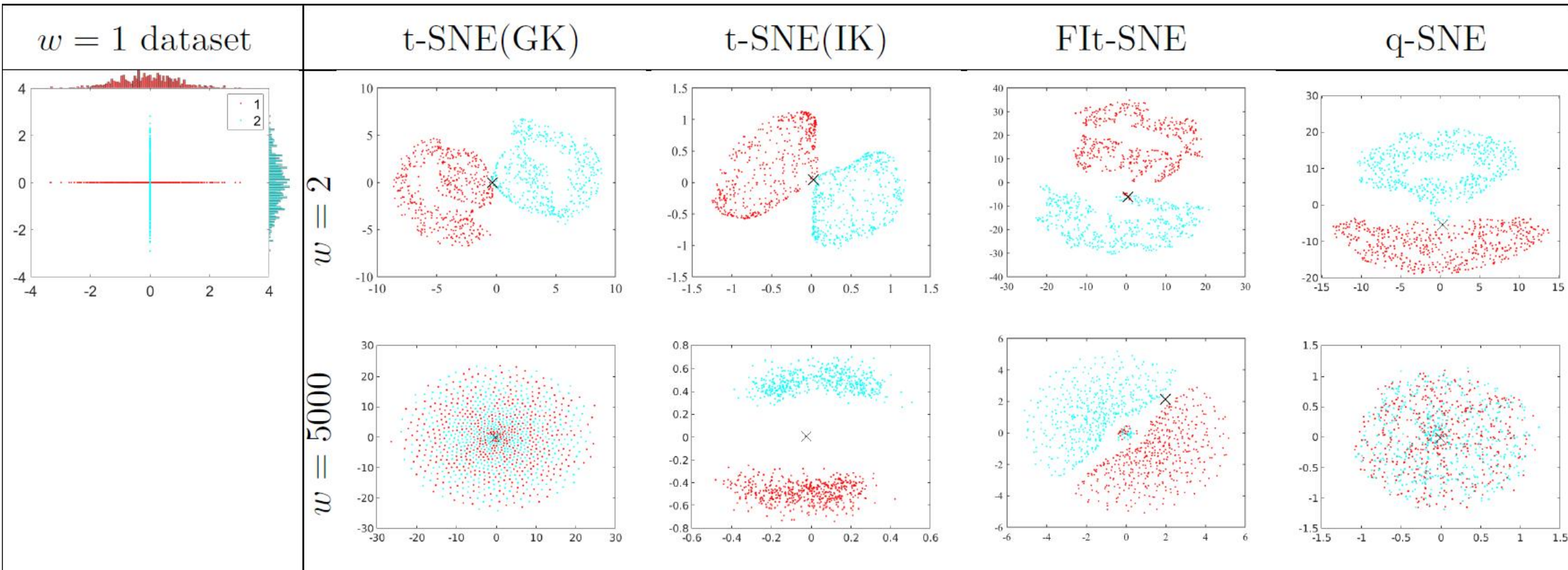
Distinguishability or Indistinguishability of a kernel

- Using the notion of (in)**distinguishability** to examine whether the feature map of a kernel is able to distinguish two distinct points in high dimensions, we analyze this ability of a metric-based Lipschitz continuous kernel as well as that of the recently introduced Isolation Kernel.
- Between the two kernels, we show that only Isolation Kernel has **distinguishability** and it performs consistently well in four tasks in both low and high dimensions, compared with distance, Gaussian and three other existing kernels :
 - indexed search for exact nearest neighbor search,
 - anomaly detection using kernel density estimation,
 - t-SNE visualization,
 - SVM classification.

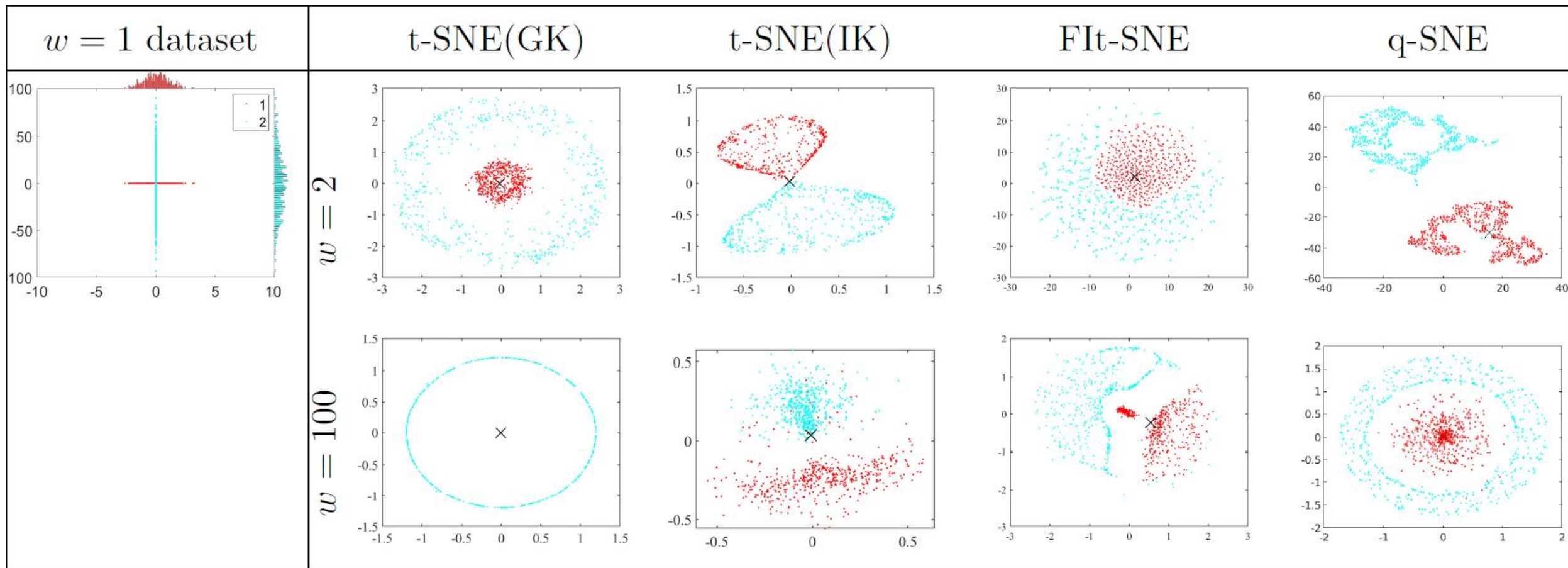
Intuition

- The key to distinguishability is the use of **space partitioning** as the means to compute the similarity. This enables points which fall into a same partition to be differentiated from points which fall outside the partition. In addition, two distinct points (even though they may fall into a same partition in some partitionings out of many) are distinguishable in the IK feature space, given a sufficient number of partitionings.
- A metric-based Lipschitz continuous kernel employs no such partitioning to determine its similarity, but solely relies on Euclidean distance in input space. The distance calculation in its feature space (or in the input space) is the root cause of its indistinguishability.

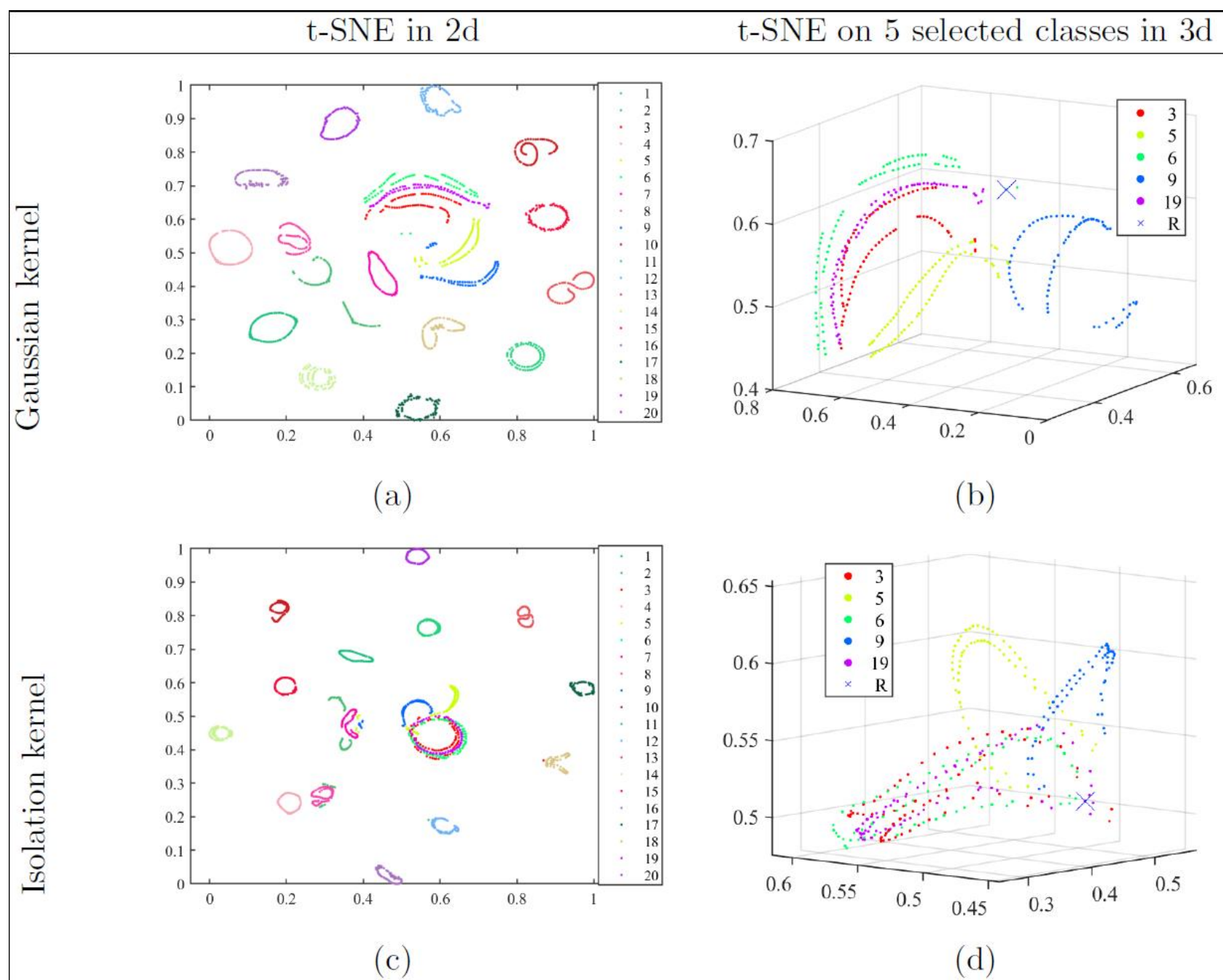
The impact on t-SNE (1)



The impact on t-SNE (2)



t-SNE Visualisation Results on COIL20 (1024 dimensions)



Discussion (high dimensions)

1. What can be expected when Euclidean distance or Cosine similarity is used to perform retrieval on a high-dimensional database?
2. What can be done instead and why?
3. Summarize what you have learned about t-SNE